

DW_cmp_dx

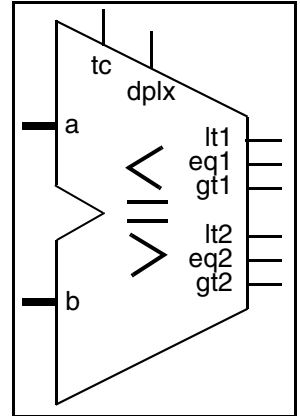
Duplex Comparator

Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- Selectable single full width Compare, or two smaller width Compare operations (duplex)
- Selectable number system (unsigned or two's complement)
- Parameterized full word width
- Parameterized partial word width (allowing for asymmetric partial width operations)
- Separate flags for Less Than, Equal To, and Greater Than
- Two sets of flags for duplex operation

Revision History



Description

The DW_cmp_dx compares operands a and b as either:

- A single comparison of *width* bits (simplex mode [*dplx* is 0]), or
- Two comparisons: one of *p1_width* (least significant) bits and one of (*width* – *p1_width*) (most significant) bits (duplex mode [*dplx* is 1]).

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>width</i> bits	Input	Input data
b	<i>width</i> bits	Input	Input data
tc	1 bit	Input	Two's complement control
dplx	1 bit	Input	Duplex mode select (active high)
lt1	1 bit	Output	Part1 : less-than output condition
eq1	1 bit	Output	Part1 : equal output condition
gt1	1 bit	Output	Part1 : greater-than output condition
lt2	1 bit	Output	Full width or part2 : less-than output condition
eq2	1 bit	Output	Full width or part2 : equal output condition
gt2	1 bit	Output	Full width or part2 : greater-than output condition

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 4	Word width of a and b
p1_width	2 to $width-2$	Word width of part1 of duplex compare

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
rpl	Ripple carry synthesis model	DesignWare
bk	Brent-Kung synthesis model	DesignWare

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW01.DW_CMP_DX_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW_cmp_dx_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_cmp_dx.v	Verilog simulation model source code

Block Diagram

The control signal tc determines whether the input data is interpreted as unsigned (tc is 0) or signed (tc is 1) numbers. Figure 1-1 shows a functional block diagram of DW_cmp_dx.

Figure 1-1 Functional Block Diagram

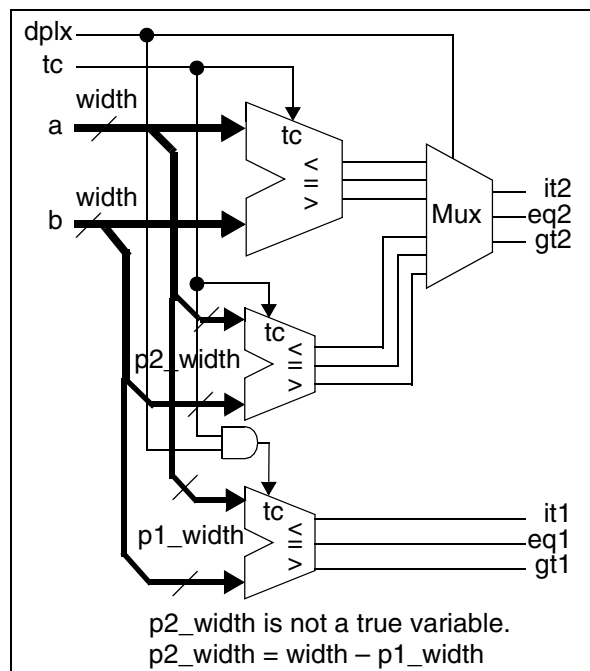


Table 1-5 Operating Modes

dplx	tc	Function
0	0	Simplex unsigned compare operation: ^a ■ It2 is 1 when a is less than b ■ eq2 is 1 when a is equal to b ■ gt2 is 1 when a is greater than b
0	1	Simplex signed compare operation: ^a ■ It2 is 1 when a is less than b ■ eq2 is 1 when a is equal to b ■ gt2 is 1 when a is greater than b

Table 1-5 Operating Modes (Continued)

dplx	tc	Function
1	0	Duplex unsigned compare operation: ■ lt2 is 1 when part2 a is less than part2 b ■ eq2 is 1 when part2 a is equal to part2 b ■ gt2 is 1 when part2 a is greater than part2 b ■ lt1 is 1 when part1 a is less than part1 b ■ eq1 is 1 when part1 a is equal to part1 b ■ gt1 is 1 when part1 a is greater than part1 b
1	1	Duplex signed compare operation: ■ lt2 is 1 when part2 a is less than part2 b ■ eq2 is 1 when part2 a is equal to part2 b ■ gt2 is 1 when part2 a is greater than part2 b ■ lt1 is 1 when part1 a is less than part1 b ■ eq1 is 1 when part1 a is equal to part1 b ■ gt1 is 1 when part1 a is greater than part1 b

a. The output values of lt1, eq1, and gt1 in the simplex modes are “don’t care.”

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_foundation_comp.all;

entity DW_cmp_dx_inst is
  generic ( inst_width      : NATURAL := 24;
            inst_p1_width  : NATURAL := 16 );
  port ( inst_a      : in std_logic_vector(inst_width-1 downto 0);
        inst_b      : in std_logic_vector(inst_width-1 downto 0);
        inst_tc      : in std_logic;
        inst_dplx    : in std_logic;
        lt1_inst     : out std_logic;
        eq1_inst     : out std_logic;
        gt1_inst     : out std_logic;
        lt2_inst     : out std_logic;
        eq2_inst     : out std_logic;
        gt2_inst     : out std_logic );
end DW_cmp_dx_inst;

architecture inst of DW_cmp_dx_inst is
begin

  -- Instance of DW_cmp_dx
  U1 : DW_cmp_dx
    generic map ( width => inst_width, p1_width => inst_p1_width )
    port map ( a => inst_a, b => inst_b, tc => inst_tc, dplx => inst_dplx,
              lt1 => lt1_inst, eq1 => eq1_inst, gt1 => gt1_inst,
              lt2 => lt2_inst, eq2 => eq2_inst, gt2 => gt2_inst );
end inst;

-- pragma translate_off
configuration DW_cmp_dx_inst_cfg_inst of DW_cmp_dx_inst is
  for inst
    end for; -- inst
end DW_cmp_dx_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```
module DW_cmp_dx_inst( inst_a, inst_b, inst_tc, inst_dplx, lt1_inst,
                      eq1_inst, gt1_inst, lt2_inst, eq2_inst, gt2_inst );

    parameter width = 24;
    parameter p1_width = 16;

    input [width-1 : 0] inst_a;
    input [width-1 : 0] inst_b;
    input inst_tc;
    input inst_dplx;
    output lt1_inst;
    output eq1_inst;
    output gt1_inst;
    output lt2_inst;
    output eq2_inst;
    output gt2_inst;

    // Instance of DW_cmp_dx
    DW_cmp_dx #(width, p1_width)
        U1 ( .a(inst_a), .b(inst_b), .tc(inst_tc), .dplx(inst_dplx),
            .lt1(lt1_inst), .eq1(eq1_inst), .gt1(gt1_inst),
            .lt2(lt2_inst), .eq2(eq2_inst), .gt2(gt2_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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